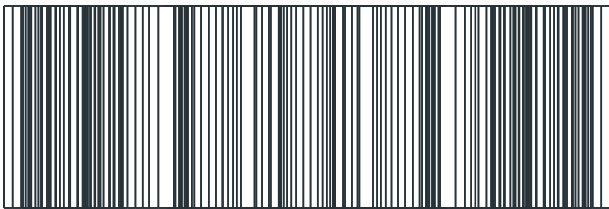
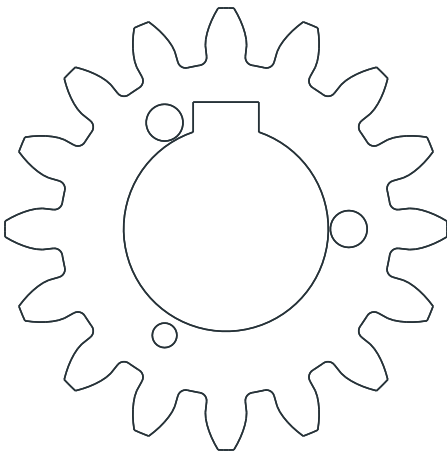


MFG-DRV-03 • v0.8 MANUFACTURING DRAWING

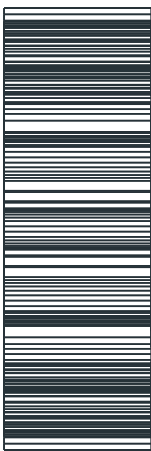
TOP X-Y



FRONT X-Z



SIDE Y-Z



Drawing MFG-DRV-03 | Part DRV-03 — M3 Z16 solid phase gear left | Qty 1

Rev final-design-fabrication-closure-v0.8 | Units mm | Scale NTS, dimensions govern | Third-angle projection | source commit 5008ee78988e17bcd38904df27b48af28fd8c594

Material: 18 mm S45C | Process/finish: wire-EDM profile/bore/keyway + finish

Overall: X 53.996 × Y 18.000 × Z 53.996 mm | general tolerance ISO 2768-m unless noted

Critical: M3 Z16 20 degree; face18; assembled pair tangential backlash 0.120-0.140 over one full mesh rotation at 20 +/-2 C; bore Ø25.01 +0.01/0; matched 8.005-8.010 keyway at datum0 deg; root fillet R1 minimum; 2xM4 clearance + 1xØ3 H7 inspection/clocking holes; full34 N.m CalculiX screen GD&T/datums: A=shaft bore axis; B=stack face; C=key/dowel angular datum; verify datum marking on supplier inspection report

Inspection: CMM/caliper against STEP and drawing; deburr and visual crack/warp inspection; CMM centre distance; blue-check backlash/contact; verify keyed torque path

Edges: remove burrs; break unspecified sharp edges C0.3–0.5. No unapproved material/tolerance substitution.

Artifact check: PASS only; fabrication HOLD pending full v0.8 gates. Physical NOT_RUN; machining USER_APPROVAL_REQUIRED.